

1. Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two normed spaces over the field $\mathbb{R}$ and Ω be a non-empty open subset of V .
 - (a) Define what is meant by the statement that the mapping $f : \Omega \rightarrow W$ is Gâteaux differentiable at $x_0 \in \Omega$ and define the Gâteaux derivative G_{x_0} of the function f at x_0 .
 - (b) Define what is meant by the statement that the mapping $f : \Omega \rightarrow W$ is Fréchet differentiable at $x_0 \in \Omega$ and define the Fréchet derivative $Df(x_0)$ of the function f at x_0 .
 - (c) Show that if $f : \Omega \rightarrow W$ is Fréchet differentiable at x_0 with Fréchet derivative $Df(x_0)$, then f is also Gâteaux differentiable at x_0 with Gâteaux derivative $G_{x_0} = Df(x_0)$.
 - (d) Show that if f is Fréchet differentiable at x_0 , then its Fréchet derivative $Df(x_0)$ at x_0 is unique.
2. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) := \begin{cases} \frac{y^3}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

- (i) Find the directional derivative $f_v(0, 0)$ of the function f at $(0, 0)$ in the direction of the vector $v = (v_1, v_2) \neq (0, 0)$.

We want to find the limit

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{f((0, 0) + t(v_1, v_2)) - f(0, 0)}{t} &= \lim_{t \rightarrow 0} \frac{f(tv_1, tv_2) - f(0, 0)}{t} \\ &= \lim_{t \rightarrow 0} \frac{t^3 v_2^3}{t^3(v_1^2 + v_2^2)} = \frac{v_2^3}{v_1^2 + v_2^2}. \end{aligned}$$

Hence,

$$f_{\vec{v}}(0, 0) = \frac{v_2^3}{v_1^2 + v_2^2}.$$

- (ii) Deduce that the function f is not Fréchet differentiable at the point $(0, 0)$.

After having obtained $f_{\vec{v}}(0, 0)$, we only have to check whether the mapping $\vec{v} \rightarrow f_{\vec{v}}(0, 0)$ is linear and continuous.

For the linearity of the above-mentioned mapping, the scalar multiplication property is trivial. From the form of $f_{\vec{v}}(0, 0)$ one would suspect that the addition does not work.

In this case, choose simple combinations of vectors like $\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$ for which we get

$$f_{\vec{i}} = 0, \quad f_{\vec{j}}(0, 1) = 1 \quad \text{and} \quad f_{\vec{i}+\vec{j}}(0, 0) = \frac{1}{2} \neq 1 = f_{\vec{i}}(0, 0) + f_{\vec{j}}(0, 0).$$

So, the function f is not Gâteaux differentiable at $(0, 0)$ and hence, is not Fréchet differentiable at that point.

3. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) := \begin{cases} \frac{\sin(\sqrt{x^2 + y^2})}{\sqrt{x^2 + y^2}} & \text{if } (x, y) \neq (0, 0), \\ 1 & \text{if } (x, y) = (0, 0). \end{cases}$$

(a) Show that f is continuous at $(0, 0)$.

This is 2AC!! We want to check whether $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = f(0, 0)$. Using polar coordinates, we want to check whether

$$\lim_{r \rightarrow 0} f(r \cos \theta, \sin \theta) = 1 \quad (\text{uniformly in } \theta) \quad \Leftrightarrow \quad \lim_{r \rightarrow 0} \sup_{\theta} |f(r \cos \theta, \sin \theta) - 1| = 0.$$

We find that

$$\lim_{r \rightarrow 0} \sup_{\theta} |f(r \cos \theta, \sin \theta) - 1| = \lim_{r \rightarrow 0} \left| \frac{\sin r}{r} - 1 \right| = 0.$$

Thus, f is continuous at the point $(0, 0)$.

(b) Establish whether the function f is Fréchet differentiable at $(0, 0)$ and find (if possible) $Df(0, 0)$ (the Fréchet derivative at $(0, 0)$).

From the definition of the Fréchet differentiability, we first need to check whether f has partial derivatives at the point $(0, 0)$.

For the partial derivative w.r.t. x we need the limit

$$\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{\frac{\sin(|x|)}{|x|} - 1}{x} = \lim_{x \rightarrow 0} \frac{\sin(|x|) - |x|}{x|x|}.$$

Now, we use the Taylor formula for the function $\sin(x)$ and round $x = 0$ to get

$$\sin(t) \approx t - \frac{1}{3!}t^3 \quad \text{for } |t| \text{ small.}$$

Thus, we get

$$\frac{\sin(|x|) - |x|}{x|x|} \approx \frac{\frac{1}{6}|x|^3}{x|x|} = \frac{1}{6}x \quad \text{for } |x| \text{ small} \quad \Rightarrow \quad \lim_{x \rightarrow 0} \frac{\sin(|x|) - |x|}{x|x|} = \lim_{x \rightarrow 0} \frac{1}{6}x = 0.$$

Thus, we get that $f_x(0, 0) = 0$ and by symmetry, also $f_y(0, 0) = 0$.

So, if f is Fréchet differentiable at $(0, 0)$, then its Fréchet derivative is $Df(0, 0) \equiv 0$.

Thus, it remains to check whether

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - f(0, 0) - 0}{\sqrt{x^2 + y^2}} = 0.$$

Notice that

$$\begin{aligned} \frac{f(x, y) - f(0, 0) - 0}{\sqrt{x^2 + y^2}} &= \frac{\sin(r) - r}{r^2} \approx \frac{1}{6}r \quad \text{as } r = \sqrt{x^2 + y^2} \rightarrow 0 \\ \Rightarrow \lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y) - f(0, 0) - 0}{\sqrt{x^2 + y^2}} &= 0. \end{aligned}$$

Therefore, the function f is Fréchet differentiable at $(0, 0)$ with Fréchet derivative $Df(0, 0) \equiv 0$.

4. Let $\mathbb{R}_{\leq n}[X]$ be the vector space of polynomials of degree less or equal to n and let

$$V := \{P|_{[0,1]} : P \in \mathbb{R}_{\leq n}[X]\}$$

be the vector space of restrictions to $[0, 1]$ of polynomials of degree less or equal to n , equipped with norm $\|\cdot\|_{\infty}$ defined by

$$\|P\|_{\infty} := \sup_{t \in [0,1]} |P(t)| \quad \forall P \in V.$$

Study the Fréchet differentiability of the mappings $f_1, f_2 : V \rightarrow \mathbb{R}$ defined by

$$f_1(P) = \int_0^1 [(P(x))^2 - 2(P(x))^3] dx \quad \text{and} \quad f_2(P) = \int_0^1 (P(x))^3 dx \quad \forall P \in V$$

and find their Fréchet derivatives $Df_1(P)$ and $Df_2(P)$ at every $P \in V$.

We first need to identify the candidates as $Df_1(P)$ and $Df_2(P)$. For this, we need to expand $f_i(P + H) - f_i(P)$, $i = 1, 2$. Notice that,

$$f_1(P + H) = \int_0^1 [(P(x) + H(x))^2] dx = \int_0^1 [(P(x))^2 + 2P(x)H(x) + (H(x))^2] dx.$$

Hence,

$$f_1(P + H) - f_1(P) = \underbrace{\int_0^1 2P(x)H(x) dx}_{\text{Candidate as } Df_1(P)(H)} + \int_0^1 (H(x))^2 dx$$

It remains to check whether,

$$\lim_{\|H\|_{\infty} \rightarrow 0} \frac{f_1(P + H) - f_1(P) - \int_0^1 [2P(x)H(x)] dx}{\|H\|_{\infty}} = 0.$$

Notice that

$$\begin{aligned} &\left| \frac{f(P + H) - f(P) - \int_0^1 [2P(x)H(x)] dx}{\|H\|_{\infty}} \right| = \frac{1}{\|H\|_{\infty}} \left| \int_0^1 [(H(x))^2] dx \right| \\ &\leq \frac{1}{\|H\|_{\infty}} \|H\|_{\infty}^2 = \|H\|_{\infty} \rightarrow 0. \end{aligned}$$

Therefore,

$$Df_1(P)(H) := \int_0^1 2P(x)H(x)dx \quad \forall P, H \in V.$$

Similarly,

$$\begin{aligned} f_2(P+H) &= \int_0^1 (P(x) + H(x))^3 dx \\ &= \int_0^1 [P(x)^3 + 3P(x)^2H(x) + 3P(x)H^2(x) + H(x)^3] dx \end{aligned}$$

Hence,

$$f_2(P+H) - f_2(P) = \underbrace{\int_0^1 3P(x)^2H(x)dx}_{\text{Candidate as } Df_2(P)(H)} + \int_0^1 [3P(x)H(x)^2 + H(x)^3] dx.$$

Hence,

$$\begin{aligned} & \left| \frac{f_2(P+H) - f_2(P) - \int_0^1 [3(P(x))^2H(x)] dx}{\|H\|_\infty} \right| \\ &= \frac{1}{\|H\|_\infty} \left| \int_0^1 [(3P(x)(H(x))^2 + (H(x))^3)] dx \right| \\ &\leq \frac{1}{\|H\|_\infty} \int_0^1 [3|P(x)|H(x)^2 + |H(x)|^3] dx \\ &\leq \frac{3\|P\|_\infty\|H\|_\infty^2 + \|H\|_\infty^3}{\|H\|_\infty} = 3\|P\|_\infty\|H\|_\infty + \|H\|_\infty^2 \rightarrow 0. \end{aligned}$$

Hence, we find by the squeeze theorem,

$$\lim_{\|H\|_\infty \rightarrow 0} \frac{f(P+H) - f(P) - \int_0^1 [3(P(x))^2H(x)] dx}{\|H\|_\infty} = 0.$$

Therefore, the function f_2 is Fréchet differentiable at every $P \in V$ with

$$Df_2(P)(H) = \int_0^1 [3(P(x))^2H(x)] dx \quad \forall P, H \in V.$$